

TIMARU, New Zealand

WMO: 937730

Lat: 44.3050S

Lon: 171.2214E

Elev: 89

StdP: 14.65

Time Zone: 12.00 (NZL)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 26.5	(c) 28.3	(d) 22.2	(e) 16.7	(f) 35.0	(g) 24.0	(h) 18.3	(i) 35.1	(j) 23.1	(k) 49.6	(l) 19.7	(m) 51.4	(n) 4.2	(o) 310	(p) 0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	80.1	62.7	76.2	61.5	72.7	60.5	65.8	74.9	64.2	72.3	62.5	69.7	11.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.6	85.1	69.2	60.9	80.1	67.6	59.4	75.8	65.7	30.7	75.1	29.4	72.4	28.2	69.9	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.0	18.2	14.9		23.0	90.1	2.1	4.8	21.4	93.6	20.2	96.4	19.0	99.1	17.5	102.6	
			WB	21.9	70.1	2.2	2.1	20.3	71.6	19.0	72.9	17.8	74.0	16.1	75.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	59.8	59.6	56.9	51.3	47.0	42.2	41.2	43.5	47.3	50.2	53.6	57.6
	DBStd	7.97	5.16	4.86	4.92	4.29	4.57	4.21	4.11	4.21	4.77	4.91	5.29	5.09
	HDD50	1059	0	0	4	32	116	238	274	206	109	57	20	3
	HDD65	5229	178	164	258	412	558	685	737	666	532	459	345	235
	CDD50	1348	304	269	216	70	23	3	2	5	27	64	126	239
	CDD65	43	17	12	6	0	0	0	0	0	0	0	2	6
	CDH74	587	211	131	71	6	1	0	0	0	2	14	48	103
	CDH80	124	51	36	11	0	0	0	0	0	0	1	6	19
(14) Wind	WSAvg	5.8	6.4	5.9	5.7	5.1	5.1	5.1	5.0	5.1	5.8	6.6	6.8	6.5
(15) Precipitation	PrecAvg	23.7	2.3	2.1	2.1	2.0	1.9	1.6	1.7	1.8	1.5	2.0	2.2	2.4
	PrecMax	31.6	7.2	8.1	5.0	4.4	6.9	6.5	4.8	8.1	3.7	3.6	4.3	6.5
	PrecMin	17.8	0.2	0.6	0.3	0.6	0.3	0.1	0.2	0.3	0.3	0.6	0.9	0.6
	PrecStd	3.7	1.4	1.6	0.9	1.1	1.4	1.6	1.3	1.8	0.9	0.8	0.9	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.8	85.0	81.0	73.8	69.6	62.2	61.5	64.7	71.4	76.0	80.2	82.3
		MCWB	65.2	64.3	63.6	57.9	55.5	49.5	49.3	50.3	54.5	57.5	61.3	62.6
	2%	DB	79.8	77.6	75.1	67.3	63.3	57.1	56.4	59.6	65.9	70.0	73.7	76.4
		MCWB	63.1	62.9	60.7	56.4	52.9	48.5	47.1	49.2	53.0	55.7	58.3	62.1
	5%	DB	75.0	73.1	70.7	63.4	58.9	53.4	52.5	55.6	61.4	65.4	68.9	71.6
		MCWB	62.4	62.6	60.1	55.3	51.6	47.0	45.7	47.8	50.8	54.1	57.8	60.4
	10%	DB	70.4	69.7	66.7	60.3	55.6	50.6	49.4	52.2	57.7	61.0	64.6	68.0
		MCWB	61.3	61.1	58.8	54.3	50.8	45.8	44.6	46.4	49.6	52.7	56.0	59.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	67.8	66.4	60.4	57.5	52.1	51.0	52.6	56.9	60.1	64.4	66.9
		MCDB	77.7	78.0	74.9	68.3	64.6	58.2	58.4	61.4	67.5	69.8	75.0	76.2
	2%	WB	65.6	65.4	63.4	57.9	54.6	49.7	48.6	50.3	54.1	57.6	61.1	64.4
		MCDB	74.9	73.9	71.4	64.0	59.5	54.3	53.4	57.0	63.4	66.9	69.9	73.0
	5%	WB	63.8	63.5	61.3	56.3	52.8	48.1	46.9	48.7	52.1	55.3	58.9	62.3
		MCDB	71.7	70.7	68.2	61.5	57.3	51.8	50.8	53.7	59.4	63.5	66.7	69.6
	10%	WB	62.0	61.8	59.6	54.9	51.2	46.6	45.4	47.3	50.4	53.4	56.8	60.2
		MCDB	68.7	68.2	65.1	59.4	54.8	50.1	49.0	51.2	55.9	59.7	63.5	66.3
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	17.4	18.1	18.4	17.4	17.8	18.0	17.6	19.3	19.1	18.5	17.1
		MCDBR	27.6	25.4	26.3	25.5	24.5	22.5	23.1	24.3	27.0	28.2	27.6	26.2
		MCWBR	13.7	13.3	14.2	16.0	16.2	15.7	15.7	15.9	16.0	15.9	14.8	13.7
	5% WB	MCDBR	22.3	21.5	22.1	21.8	19.8	19.0	19.8	20.3	23.7	23.9	23.3	21.9
		MCWBR	12.6	12.2	13.5	14.6	13.9	14.1	14.4	14.1	15.6	15.2	14.0	13.1
(40) Clear-Sky Solar Irradiance	taub		0.329	0.324	0.308	0.295	0.281	0.275	0.272	0.286	0.303	0.312	0.315	0.327
	taud		2.568	2.594	2.632	2.648	2.646	2.607	2.631	2.629	2.553	2.554	2.576	2.571
	Ebn at Noon		316	308	299	280	261	250	261	279	295	309	318	318
	Edh at Noon		33	30	26	22	18	17	18	22	28	31	32	33
(44) All-Sky Solar Radiation	RadAvg		1908	1706	1317	881	550	429	510	738	1131	1557	1877	1966
(45)	RadStd		154	146	65	75	36	27	37	49	70	103	118	121

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+0.50	+0.50	+0.58	N/A	N/A	N/A	-74	N/A	N/A	N/A